Meeting 2 — Feature Set, Hardware Architecture, and User Interface Design

Project: Digital Harmonium Workstation

Phase: 3.1 — Runtime Design Preparation

Meeting Objective

The objective of this meeting is to finalize the functional scope of the Digital Harmonium Workstation before designing the firmware runtime architecture.

Meeting 1 established *why* the instrument exists.

Meeting 2 establishes *what* the finished instrument will contain.

This document freezes the functional identity of the workstation. Future additions remain possible, but all future firmware architecture, task scheduling, memory allocation, and runtime design will be based on the decisions made in this meeting.

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1. Design Philosophy

The Digital Harmonium Workstation is intended to become a practical musical instrument rather than a demonstration of embedded programming.

The project is guided by the following principles:

- Restore a discarded electronic keyboard and give it a second life.
- Produce a low-cost workstation suitable for Indian Classical musicians.
- Replace several expensive standalone devices with one integrated instrument.
- Prioritize musical usability over feature quantity.
- Keep the interface simple enough for musicians of any experience level.
- Maintain a modular architecture so additional features can be added later without redesigning the entire system.

The workstation is designed primarily for musicians rather than programmers.

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2. Instrument Set

After discussion, the initial instrument collection is frozen as follows.

Included

Harmonium

Primary instrument.

The harmonium defines the identity of the entire project.

Organ

Secondary synthesized instrument.

Provides tonal variation while remaining computationally inexpensive.

Tanpura

Continuous drone accompaniment.

Supports Indian Classical practice and performance.

Tabla

Rhythmic accompaniment engine.

Instead of storing complete taals, the system stores individual tabla bols.

Examples include:

- Dha
- Dhin
- Na
- Tin
- Ta
- Ge

Each taal is constructed dynamically by sequencing these recorded bols.

Advantages:

- Extremely small storage requirements
- Easy addition of new taals
- Greater flexibility
- Better modularity

Supported taals initially include:

- Tritaal
- Ektaal
- Jhaptal
- Chautaal
- Rupak
- Dadra

Additional taals may be added later.

Deferred Instruments

The following instruments are intentionally postponed:

- Flute
- Sitar
- Strings
- Pads
- Additional voices

These instruments are not removed from the project.

They are deferred because the architecture allows new VHL instruments to be added without changing firmware.

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3. Performance Features

The workstation shall provide the following musician-oriented features.

Instrument Selection

Users can select any installed instrument.

Examples:

- Harmonium
- Organ

Future instruments will appear automatically once their VHL files are available.

Master Volume

Global output level.

Tempo Control

Tempo affects:

- Tabla
- Tanpura
- Metronome

Transpose

Instead of traditional "Patti Shift," the workstation uses a general transpose system.

Transpose changes the musical key while preserving fingering.

Tabla Controls

Dedicated controls include:

- Start
- Stop
- Tempo

Tanpura Controls

Dedicated controls include:

- Start
- Stop
- Tempo
- Transpose

Recording

One of the defining features of the workstation.

Unlike commercial keyboards that record audio, this project records musical performance events.

Recorded information includes:

- Key pressed
- Key released
- Timestamp
- Instrument
- Performance timing

Advantages:

- Very small file sizes
- Instrument-independent playback
- Tempo can change
- Transpose remains possible
- Musical information is preserved rather than audio

Playback

Playback reconstructs the original performance by replaying stored musical events.

During playback:

- Audio is reproduced.
- Keyboard LEDs illuminate corresponding keys.
- Students can visually follow the performance.

This transforms playback into a learning tool rather than merely an audio recorder.

Metronome

Integrated practice metronome.

Panic Button

A dedicated emergency function shall stop all currently playing notes.

Responsibilities include:

- Stop every active voice.
- Clear voice allocation tables.
- Remove hanging notes.

- Immediately silence audio.

This greatly simplifies debugging and improves live reliability.

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4. Educational LED System

The original keyboard LEDs become an integral educational feature.

Rather than functioning as simple indicators, they become part of the user experience.

Applications include:

Live Playing

Current note illumination.

Recording

Visual confirmation of played keys.

Playback

Automatic illumination synchronized with recorded performances.

Practice Mode

Future teaching applications.

Because the LEDs already exist in the original keyboard hardware, this feature provides significant educational value with minimal additional hardware cost.

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5. Audio Storage Strategy

A major architectural decision was finalized during this meeting.

Previous Approach

Store all instruments permanently inside ESP32 flash memory.

This approach limits future expansion.

Final Architecture

Store all VHL files on the SD card.

Example:

```
SD Card

harmonium.vhl

organ.vhl

tanpura.vhl

tabla.vhl

future instruments...
```

When the user selects an instrument:

```
User selects instrument

↓

Read corresponding VHL file

↓

Load into RAM

↓

Instrument becomes active
```

Advantages:

- Unlimited instrument expansion

- Firmware remains unchanged
- Easy updates
- Modular architecture
- Simplified maintenance

The SD card therefore becomes an instrument library rather than an audio streaming device.

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6. Display System

The workstation shall initially use a standard 16×2 LCD.

Reasons:

- Already available
- Simple
- Reliable
- Low processor overhead
- Sufficient for required information

Displayed information includes:

- Current instrument
- Tempo
- Transpose
- Recording status
- Playback status
- Menu navigation

Future migration to larger displays remains possible without major firmware redesign.

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7. Control Interface

The project intentionally avoids complicated user interfaces.

Current interface consists of:

- Existing keyboard matrix
- Dedicated function buttons
- LCD display

Although rotary encoders were considered, they are postponed because:

- Existing button matrix already exists.
- Mechanical integration would complicate enclosure design.

- Buttons sufficiently satisfy current requirements.

Future revisions may introduce rotary encoders if desired.

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8. Audio Hardware

The audio subsystem shall consist of:

ESP32

↓

MAX98357A I^2S Digital Audio Amplifier

↓

Internal Speaker

and

↓

3.5 mm Headphone Output

Using an external I^2S DAC/amplifier improves audio quality while simplifying ESP32 analog hardware requirements.

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9. Inputs

Primary inputs include:

- Keyboard matrix
- Function buttons
- SD card
- USB programming interface
- Power input

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10. Outputs

Primary outputs include:

- Internal speaker
- 3.5 mm headphone jack
- 16×2 LCD
- Keyboard LEDs

These outputs define the complete user interaction system.

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11. Modularity

Every major subsystem remains modular.

Future additions such as:

- Flute
- Sitar
- Strings
- Larger displays
- Rotary encoders
- Additional effects

can be integrated without redesigning the complete firmware.

This modularity was established as a core architectural requirement.

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12. Meeting Conclusions

Meeting 2 finalizes the complete functional identity of the Digital Harmonium Workstation.

The workstation is now defined as:

- A restored electronic keyboard.
- A low-cost Indian Classical workstation.
- A modular instrument supporting VHL-based voices.
- A system with integrated harmonium, organ, tanpura, and programmable tabla accompaniment.
- A learning-oriented instrument featuring event-based recording and LED-assisted playback.
- A platform designed for future expansion while maintaining a stable runtime architecture.

The decisions documented in this meeting establish the functional foundation upon which the complete ESP32 runtime architecture will be designed.

Meeting Status: Finalized

Next Meeting: Meeting 3 — Runtime Architecture, Boot Sequence, Object Lifetimes, Task Scheduling, Audio Pipeline, and Complete ESP32 System Lifecycle.